A Cogging Torque Suppression Algorithm Based on Model Predictive Control

Peng GAO*, Peng GONG*, Bin TANG*, Piao HE*, Jianing JI*, Guangwei ZHANG*

* National Key Laboratory of Mechatronic Engineering and Control, School of Mechatronical Engineering, Beijing Institute of Technology, Beijing, China

3120235461@bit.edu.cn, penggong@bit.edu.cn, 3120230249@bit.edu.cn, 3220230127@bit.edu.cn, jjianing2023@126.com, 6120240118@bit.edu.cn

***Abstract*—Cogging torque is a unique issue in the servo systems of permanent magnet synchronous motor (PMSM) and a key problem that must be addressed in the design and manufacturing of high-performance permanent magnet motors. Due to their inherent structural characteristics, PMSMs inevitably produce cogging torque, which leads to disturbances and noise, particularly detrimental to motor control systems operating at low speeds. Additionally, the presence of cogging torque affects the positioning accuracy of the rotor at any given moment. This paper presents a high-quality control method for suppressing cogging torque. The proposed cogging torque suppression strategy is studied through simulations and compared with traditional cogging torque suppression strategies. The simulation results indicate that the proposed high-precision positioning servo control method is effective in suppressing cogging torque.**

***Keywords*—PMSM; Cogging torque; Model predictive control; Low-speed ripple suppression; Position accuracy.**

I. INTRODUCTION

Permanent magnet synchronous motor (PMSM) inevitably generate cogging torque due to their inherent structural characteristics, leading to vibrations and noise. Researchers both domestically and internationally have proposed various optimization and improvement measures. [1] classifies and summarizes cogging torque suppression methods from a structural optimization perspective, integrating recent research findings.

Inspired by the characteristics of cogging torque, BU proposes a virtual cogging torque (VCT) control method based on a permanent magnet synchronous motor model to reduce speed fluctuations in direct-drive PMSM servo systems under low-speed conditions [2]. A comparative study was conducted on PMSM, PMSM with Circular Flux Barriers, PMSM with Tubular Flux Barriers, and Interior Permanent Magnet motors to obtain parameters such as cogging torque [3].To further reduce torque pulsations, Noguchi proposes a pulse merging method that samples an additional reference value compared to traditional methods to improve current control. This approach has shown significant effects in suppressing cogging torque in high-speed regions [4].

In addition to optimizing motor control algorithms, cogging torque suppression can also be addressed through structural optimization of the motor itself. In [5], it presents a new topology from the perspective of air gap magnetic permeability, utilizing an unequal slot width (USW) design to reduce cogging torque in V-type permanent magnet vernier (V-PMV) motors. Modifying the rotor structure can also help suppress cogging torque, as described by Cao [6], who proposed two methods for reducing cogging torque. In [7], Kanapara indicates that appropriate design modifications to permanent magnet motors can reduce cogging torque. Gao presents a method for reducing cogging torque in permanent magnet (PM) motors through dynamic control based on harmonic torque compensation (HTC) [8].

Typically, cogging torque suppression methods involve first simulating the magnitude of cogging torque through finite element analysis, followed by structural or control measures to suppress torque pulsation. In [9], this study employs the chamfering method of right angles to suppress the cogging torque in flux-switching permanent magnet (FSPM) machines, applied on both the stator and rotor sides. Sun proposes a model predictive control method for motors, incorporating an observer to balance system disturbances with a degree of advancement [10].

Altering the motor structure through design to suppress cogging torque can compromise the integrity of the motor's structure, thereby degrading its performance. In certain special operating conditions where a high instantaneous power density is required, cogging torque tends to increase alongside the motor's high instantaneous power density. Therefore, methods that alter the motor structure to suppress cogging torque are not suitable for special operating environments. This paper proposes a high-precision positioning servo control method to suppress cogging torque without altering the motor's original structural parameters.

II. PRELIMINARY AND CONCEPT

A. Concept of Cogging Torque

Cogging torque is a unique periodic torque characteristic of permanent magnet synchronous motor (PMSM) that occurs even under no-load conditions. The formation of cogging torque results from the interaction between the rotor permanent magnets and the stator slots, generating torque as the rotor tends to move in the direction of minimum magnetic reluctance. When the motor is in a no-load state, there are several stable positions that can hold the rotor in place, with the number of stable positions determined by the internal structure of the motor. When manually rotating the rotor, one can distinctly feel varying torque as the rotor turns; therefore,

ISBN: 979-11-88428-13-7

cogging torque is also referred to as positioning torque. The magnitude of cogging torque can be represented as the negative derivative of the magnetic field energy W stored in the motor with respect to the relative angular position α between the stator and rotor. This can be expressed as:

$$T_{cog} = -\frac{\partial W}{\partial \alpha} \tag{1}$$

Where T_{cog} is the cogging torque of the motor; W is the magnetic co-energy of the motor; and α\alphaα is the relative angular position between the stator and rotor. From equation (1), it can be seen that the magnitude of cogging torque varies with the relative angular position between the stator and rotor, resulting in a torque that is not uniform. As the relative position of the rotor magnetic poles and slots changes, cogging torque arises, resisting the rotor's motion. The cogging torque generated by all slots in the motor and the rotor magnetic poles is summed to yield the total cogging torque of the motor.

The cogging torque of the motor can be suppressed by changing structural design and optimizing control algorithms. In terms of structural design, this can involve modifying the stator structure, changing the rotor structure, and optimizing winding design. However, for motors that have already been manufactured, employing structural optimization to suppress cogging torque can significantly increase costs and may also compromise the motor's structure, reducing its operational performance. Therefore, optimizing the motor's control algorithms can not only mitigate the effects of cogging torque but also lower costs while preserving the motor's original performance characteristics.

B. PI-based Motion Control Strategy

To facilitate the study of the mathematical model of the permanent magnet synchronous motor, the following assumptions are made:

1) Ignoring eddy currents, hysteresis, and core losses;

2) The spatial magnetic field exhibits a sinusoidal distribution in space;

3) The stator windings are uniformly and symmetrically distributed.

To facilitate the design of the controller, i_d, i_q and ω are chosen as state variables. Therefore, the mathematical model of the permanent magnet synchronous motor can be expressed as:

$$\begin{cases} \dfrac{di_d}{dt} = -\dfrac{R_S i_d}{L_d} + \dfrac{u_d}{L_d} + n_p i_q \omega \\ \dfrac{di_q}{dt} = -\dfrac{R_S i_q}{L_q} + \dfrac{u_q}{L_q} - n_p i_d \omega - \dfrac{n_p \phi_v \omega}{L_q} \end{cases} \tag{2}$$

The dynamic equation of the PMSM is:

$$\frac{d\omega}{dt} = \frac{K_t i_q}{J} - \frac{B\omega}{J} - \frac{T_L}{J} \tag{3}$$

Where ω is the rotor speed; i_d and i_q are the stator currents in the *d-q* axes; u_d and u_q are the stator voltages in the *d-q* axes; L_d and L_q are the stator inductances in the *d-q* axes, satisfying L_d=L_q=L; R_s is the stator resistance; φ_d and φ_q are the magnetic fluxes in the *d-q* axes; φ_v is the rotor magnetic flux; n_p is the number of pole pairs; J is the moment of inertia; and T_L is the load torque; B is the viscous friction coefficient; K_t=$3n_p\varphi_v/2$.

As shown in Fig.1, the diagram illustrates the principle of the PMSM drive control system. The drive control system consists of three controllers, space vector pulse width modulation (SVPWM), an inverter, a field-oriented control device, and the PMSM. The three controllers include one speed loop controller and two current controllers, employing a traditional cascaded control loop structure. Among them, the current loop controllers include two PI controllers, which are used to stabilize the current tracking errors in the *d-q* axes.

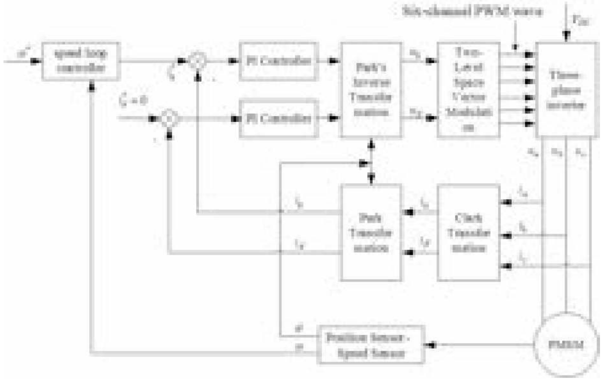


Figure 1. Schematic diagram of PMSM drive control system

From the above diagram, it can be seen that the rotor position θ and the rotor angular velocity ω can be obtained through position and speed sensors. The measured currents i_a and i_b are transformed into i_d and i_q using Clarke and Park transformations. The current i_d is set to zero to maintain constant flux operating conditions. The reference current i_q is output by the speed loop controller. Therefore, the control method for suppressing cogging torque needs to be implemented in the speed loop controller, which involves designing a speed loop controller that suppresses cogging torque.

C. MPC+ESO-based Motion Control Strategy

The principle of Model Predictive Control (MPC) is straightforward and can meet the demands of multi-objective control, featuring a rapid dynamic response. It is particularly suitable for the nonlinear and multi-coupled complex systems of permanent magnet synchronous motors.

Designing a speed loop controller based on Model Predictive Control for the motor achieves better suppression of cogging torque. Model Predictive Control can predict the future outputs of the system based on current system information and compensate for model mismatches, noise, and disturbances occurring in each sampling period of the system.

The forward Euler method is chosen to discretise the equations. The forward Euler method can be expressed as:

$$y_{n+1} = y_n + hf(y_n, t_n) \tag{4}$$

In addition, there is an initial condition:

$$y(t_0) = y_0 \tag{5}$$

ISBN: 979-11-88428-13-7

Using the forward Euler method to discretise the dynamic equations for cogging torque suppression in the PMSM results in the following prediction equation for the rotor speed of the PMSM:

$$\omega(k+1) = \omega(k) + \frac{T_c}{J_m}\left[T_e(k) - T_{cog}(k)/N_s\right] \quad (6)$$

Where T_c is the sampling period; $\omega(k)$ is the rotor speed at time k; $\omega(k+1)$ is the predicted rotor speed at time $k+1$; T_e is the electromagnetic torque of the PMSM; and T_{cog} is the cogging torque.

From the equation (6), the predicted rotor speed of the permanent magnet synchronous motor for each step can be obtained. Therefore, when the prediction horizon is $(k+i)_{th}$, the predicted speed $\omega_m(k+1)$ is:

$$\omega_m(k+i) = \omega_m(k) + \frac{iT_c}{J_m}\left(K_T i_q^*(k) - T_{cog}(k)/N_s\right) \quad (7)$$

As shown in Fig.2, this illustrates the principle of the MPC+ESO speed loop controller.

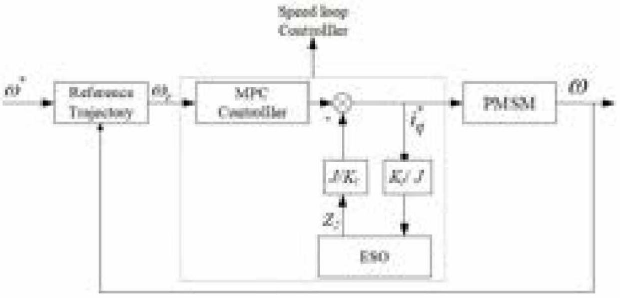


Figure 2. Schematic Diagram of the MPC+ESO Speed Loop Controller

From the Fig.2, it can be concluded that the output of the composite controller drives the motor to produce ω. Therefore, the dynamic equation of the motor can be expressed as:

$$\begin{aligned}\dot{\omega} &= \frac{K_t}{J} i_q - \frac{B}{J}\omega - \frac{T_L}{J} \\ &= \frac{K_t}{J} i_q^* - \frac{B}{J}\omega - \frac{T_L}{J} - \frac{K_t}{J}(i_q^* - i_q) \\ &= \frac{K_t}{J} i_q^* + d(t)\end{aligned} \quad (8)$$

Where $d(t)=-(B\omega/J)-(T_L/J)-(K_t/J)(i_q^*-i_q)$ represents the total disturbance in the controller, including friction, external load disturbances, and in this paper, the disturbances caused by cogging torque.

Defining $x_2=d(t)$ and $x_1=\omega$, Equation (15) can be expressed as a system of state equations:

$$\begin{cases}\dot{x}_1 = x_2 + \frac{K_t}{J} i_q^* \\ \dot{x}_2 = c(t)\end{cases} \quad (9)$$

Where $c(t)$ represents the rate of change of the system's uncertainty and total disturbance $d(t)$. It can then be expressed as a second-order linear ESO system:

$$\begin{cases}\dot{z}_1 = z_2 - 2p(z_1 - x_1) + b_0 i_q^* \\ \dot{z}_2 = -p^2(z_1 - x_1)\end{cases} \quad (10)$$

Where $-p$ represents the poles of the Extended State Observer (ESO), with $p>0$; z1 is the estimated value of the speed ω; z_2 is the estimated value of the total disturbance $d(t)$ in the system; and b_0 is the estimated value of K_t/J. Based on the previous analysis, it follows that $z_1(t) \rightarrow \omega$ and $z_2(t) \rightarrow a(t)$. Utilizing this information, disturbance compensation can be achieved.

III. Simulation Validation

A. Parameter Configuration

The input analog value at the motor load end is set to 1 N·m, simulating a sinusoidal distribution of cogging torque. No additional load is introduced, simulating the motor under no-load conditions. Through simulation analysis, the effectiveness of traditional PI control in suppressing cogging torque is evaluated. To enhance comparability, an ESO is introduced to form a reference group for comparing the control effects of PI and PI+ESO. Subsequently, the proposed cogging torque suppression algorithm based on MPC+ESO is subjected to simulation analysis, comparing the control effects of MPC and MPC+ESO.

The parameter settings for the motor are as follows: the rated voltage of the motor is 48VDC; the rated current is 23A; the rated power is 1.2kW; the instantaneous maximum current is 92A; the magnetic flux is 0.11Wb; the line inductance is 0.6mH; the moment of inertia is 0.069kg·m²; the Magnetic Damping Coefficient is 0.02; and the stator resistance is 0.1Ω.The simulation study in this paper is divided into two parts: positioning accuracy testing and speed ripple suppression testing.

B. Positioning Accuracy Testing

Fig.3 shows the positioning accuracy simulation results when the input commanded position is 180°, comparing the speed loop PI control and the speed loop PI+ESO control methods.Fig.3(a) shows the position curve based on speed loop PI control when the commanded position is 180°. It can be observed that when the rotor position converges to the commanded position, there is an overshoot of approximately 0.4°, and oscillations occur during the convergence process.

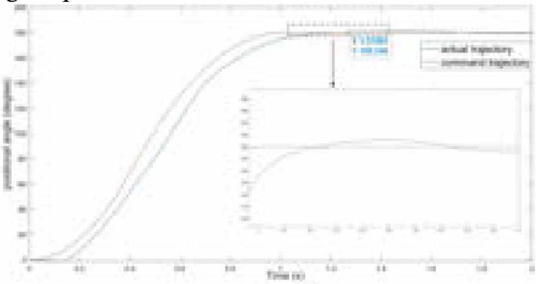
In Fig.3(b), it can be seen that the actual rotor position curve shows significant improvement compared to the PI control. However, there are still slight oscillations during the convergence process, with a maximum overshoot of about 0.025°. Ultimately, the rotor position gradually approaches the commanded position with a certain error.

Fig.4 shows the positioning accuracy simulation results based on speed loop MPC and speed loop MPC+ESO when the commanded position is 180°. From Fig.4(a), it can be observed that the cogging torque suppression algorithm based on speed loop MPC has a significant suppressive effect on the cogging torque generated by the elastomer-driven motor, demonstrating a notable improvement compared to the previous simulation results based on traditional PI control.

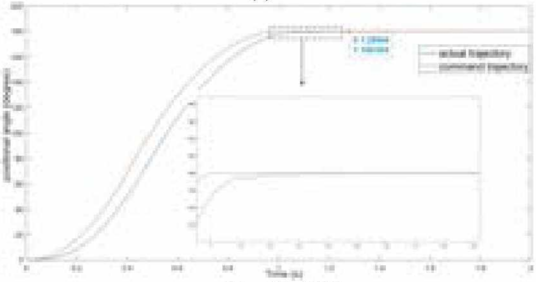
From Fig.4(b), it can be seen that the balanced steady state of the entire control system has been further improved, resulting in a shorter time for the rotor to reach the specified

ISBN: 979-11-88428-13-7

position. According to the simulation results, after entering the balanced steady state, the positioning accuracy error of the rotor is 0.02°, indicating that the rotor has reached the commanded position accurately under ideal conditions without any error. This validates the effectiveness of the proposed MPC+ESO control algorithm in suppressing cogging torque.

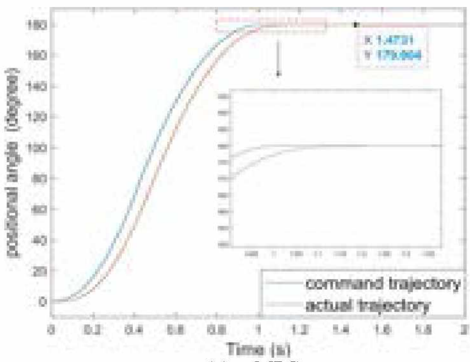


(a) PI

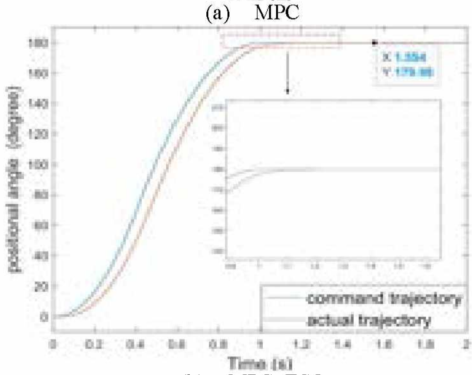


(b) PI+ESO

Figure 3. Positioning Accuracy Simulation Results Based on Speed Loop PI and Speed Loop PI+ESO.



(a) MPC

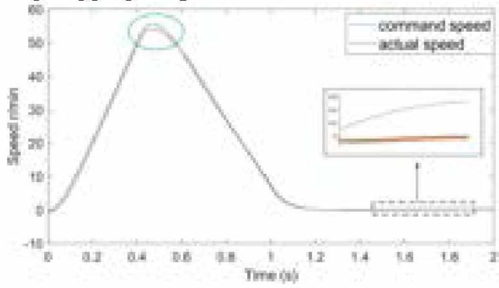


(b) MPC+ESO

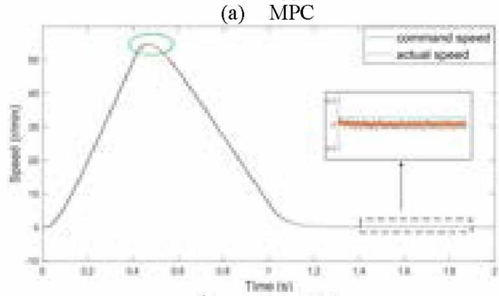
Figure 4. Positioning Accuracy Simulation Results Based on Speed Loop MPC and Speed Loop MPC+ESO.

As shown in Fig.5, the positioning accuracy speed simulation results based on speed loop MPC and speed loop MPC+ESO are illustrated. In Fig.5(a), there remains an issue where the actual speed does not keep up with the commanded speed. The error between the actual speed and the commanded speed is quite large at the peak in the figure. From Fig.5(b), it can be seen that the cogging torque suppression algorithm based on MPC+ESO has improved the stability of the elastomer-driven motor control system. The error between the actual speed and the commanded speed has significantly decreased compared to Fig.5(a).

Comparing the results shown in Fig.5, it can be seen that the MPC+ESO control algorithm achieves better control over speed errors, leading to a more stable rotor during rotation. The comparison of simulation results validates the effectiveness of the proposed MPC+ESO control algorithm in suppressing cogging torque.



(a) MPC



(b) MPC+ESO

Figure 5. Speed Simulation Results for Positioning Accuracy Based on Speed Loop MPC and Speed Loop MPC+ESO.

C. Speed ripple Suppression

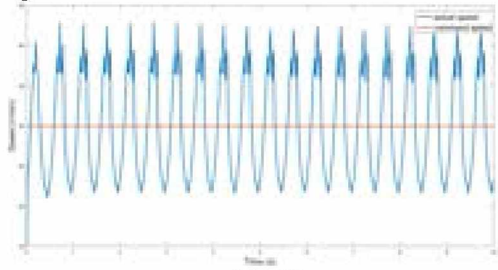
Fig.6 shows the speed fluctuation suppression simulation results based on speed loop PI and speed loop PI+ESO when the commanded speed input is 30 r/min. Fig.6(a) illustrates the speed curve when the specified speed is 30r/min using the speed loop PI control method. It can be seen that the actual speed fluctuates significantly, with an error of approximately ±13r/min between the actual speed and the commanded speed. The error is substantial, and there is also a noticeable periodicity in the cogging torque, resulting in periodic speed ripples.

Fig.6(b) shows the speed curve when the specified speed is 30r/min, using the speed loop PI+ESO. With the introduction of the ESO, it is clear that the fluctuations in actual speed are reduced. Additionally, the periodic speed fluctuations caused by cogging torque have been effectively suppressed. The error between the actual speed and the commanded speed is

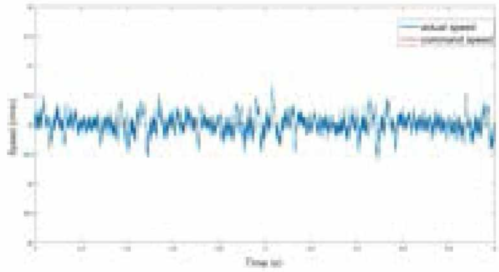
ISBN: 979-11-88428-13-7

approximately ±0.5/min. The suppression algorithm for cogging torque still requires improvement.

Fig.7 shows the speed fluctuation suppression simulation results based on speed loop MPC and speed loop MPC+ESO when the commanded speed input is 10r/min. In Fig.7 (a), it can be observed that the cogging torque suppression algorithm based on speed loop MPC significantly reduces the speed fluctuations caused by cogging torque. Under ideal conditions, the amplitude of the speed fluctuations is approximately 10.6 r/min. After introducing the ESO, the periodic fluctuations caused by cogging torque are notably suppressed, and the amplitude of the speed fluctuations is significantly reduced. Combined with the two sets of specified speed simulations shown in Fig.7, this validates the effectiveness of the proposed speed loop MPC+ESO control strategy in suppressing cogging torque in PMSM, as well as the reliability of the proposed control algorithm.

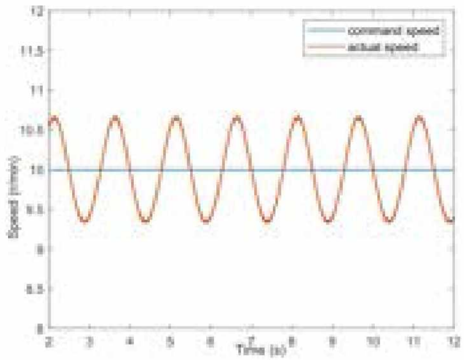


(a) PI

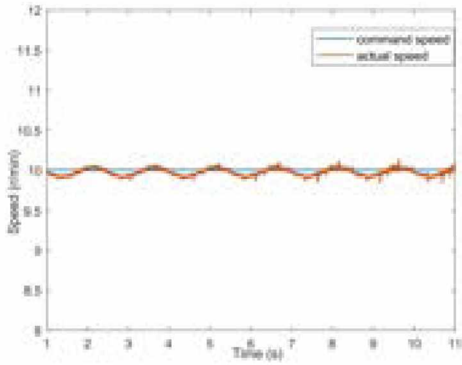


(b) PI+ESO

Figure 6. Speed Ripple Suppression Simulation Results Based on Speed Loop PI and Speed Loop PI+ESO.



(a) MPC



(b) MPC+ESO

Figure 7. Speed Ripple Suppression Simulation Results Based on Speed Loop MPC and Speed Loop MPC+ESO.

IV. CONCLUSION

This paper addresses the issues of speed fluctuations caused by cogging torque and the positioning accuracy of the motor when operating under low-speed conditions. A control method utilizing speed loop MPC+ESO is proposed to suppress the effects of cogging torque. The effectiveness of the proposed control algorithm in suppressing cogging torque is validated through simulation experiments on positioning accuracy and speed ripple suppression on the simulation platform, providing a theoretical basis for subsequent experiments.

ACKNOWLEDGMENT

This research was supported in part by National Natural Science foundation of China (No.62402045、No.62073039、No.62472037、No.62203048).

Corresponding author: Guangwei Zhang.

REFERENCES

[1] Wenbo B, Huajun R. Optimization of Cogging Torque Based on the Improved Bat Algorithm[J]. International Journal of Information Technologies and Systems Approach (IJITSA),2023,16(2):1-19..

[2] BU, FEIFEI, YANG, ZHIDA, GAO, YU, et al. "Speed Ripple Reduction of Direct-Drive PMSM Servo System at Low-Speed Operation Using Virtual Cogging Torque Control Method" [J]. IEEE Transactions on Industrial Electronics,2021,68(1):160-174.

[3] P. Sundaramoorthy, P. Devireddy, C. Thummala, S. Valmiki, P. A. Kaperla and D. Padmanabhan, "PMSM Motors in the Air: Investigating Torque Characteristics for Optimal Drone Performance," 2024 IEEE Third International Conference on Power Electronics, Intelligent Control and Energy Systems (ICPEICES), Delhi, India, 2024, pp. 286-291.

[4] S. Noguchi and H. Fujimoto, "Torque Ripple Reduction for PMSM based on PWM Pulse Merging Method for High-Speed Range," [C]. 2021 IEEE International Conference on Mechatronics (ICM), 2021, pp. 1-6.

[5] T. Wang, X. Zhu, Z. Xiang, W. Pu and L. Quan, "Cogging Torque Reduction of a V-Shaped PM Vernier Motor from Perspective of Airgap Permeance." [C]. 2020 IEEE International Conference on Applied Superconductivity and Electromagnetic Devices (ASEMD), 2020, pp. 1-2.

[6] Z. Cao and J. Liu, "Cogging Torque Reduction for Outer Rotor Interior Permanent Magnet Synchronous Motor." [C]. IECON 2020 The 46th Annual Conference of the IEEE Industrial Electronics Society, 2020, pp. 2689-2693.

[7] A. J. Kanapara and K. P. Badgujar, "Performance Improvement of Permanent Magnet Brushless DC Motor through Cogging Torque

ISBN: 979-11-88428-13-7

Reduction Techniques." [C]. 2020 21st National Power Systems Conference (NPSC), 2020, pp. 1-6.
[8] J. Gao, Z. Xiang, L. Dai, S. Huang, D. Ni and C. Yao, "Cogging Torque Dynamic Reduction Based on Harmonic Torque Counteract." [J]. IEEE Transactions on Magnetics, 2022, vol. 58, no. 2, pp. 1-5.
[9] Zhu X, Hua W, Wu Z. Cogging torque minimisation in FSPM machines by right-angle-based tooth chamfering technique[J]. IET Electric Power Applications,2018,12(5):627-634..
[10] L. Sun, X. Li, L. Chen, H. Shi and Z. Jiang, "Dual-Motor Coordination for High-Quality Servo with Transmission Backlash," in IEEE Transactions on Industrial Electronics, vol. 70, no. 2, pp. 1182-1196, Feb. 2023.

Peng Gao received the MS degree in Armament Science and Technology from Nanjing University of Science and Technology in 2023, and now he is a Ph.D. candidate in School of Mechantronical Engineering, Beijing Institute of Technology. His research interests include wireless network simulation and emulation, control science and engineering, information security, wireless communication and so on.

Peng Gong received the BS degree in Mechatronical Engineering from Beijing Institute of Technology, Beijing, China, in 2004, and the MS and Ph.D. degrees from the Inha University, Korea, in 2006 and 2010, respectively. In July 2010, he joined the School of Mechatronical Engineering, Beijing Institute of Technology, China. His research interests include link/system level performance evaluation and radio resource management in wireless systems, information security, and the next generation wireless systems such as 3GPP LTE, UWB, MIMO, Cognitive radio and so on.

Bin Tang obtained the BS degree in Electronic Science and Technology from Beijing Institute of Technology in 2023 and is currently a Master's student in the School of Mechanical and Electrical Engineering at Beijing Institute of Technology. His research interests include wireless network simulation and emulation, control science and engineering, communication algorithms, wireless communication, etc.

Piao He obtained the BS degree in Aerospace Engineering from Beijing Institute of Technology in 2023 and is currently a Master's student in the School of Mechanical and Electrical Engineering at Beijing Institute of Technology. Her research interests include wireless network simulation and emulation, control science and engineering, communication algorithms, wireless communication, etc.

Jianing Ji received the BS degree in School of Mechanical and Electrical Engineering from North University of China in 2024, and now she is a Master candidate in School of Mechatronical Engineering, Beijing Institute of Technology. Her research interests include wireless network simulation and emulation, control science and engineering, wireless communication and so on.

Guangwei Zhang received Ph.D. degree in school of Mechatronical Engineering from Beijing Institut in 2021. He is currently pursuing the postdoctor in Beijing Institute of Technology, Beijing, China. He is working radar signal processing and intelligent detection, control science and engineering, network simulation and emulation and so on.

ISBN: 979-11-88428-13-7